

Homework #2

Subheading: *Solving exercise problems on continuous random variables*

Course: *Probability and Random Variables* – Professor: *Bang Chul Jung*
Due Date: *15:00 May 13th, 2026 (before the class start)*

Problem 1

The CDF of a random variable X is given by

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-2x}, & x \geq 0. \end{cases}$$

- (a) Find the PDF $f_X(x)$.
- (b) Find $\Pr(1 < X < 2)$.
- (c) Find $\Pr(X < \mathbb{E}[X])$

Problem 2

A temperature sensor is used to measure the true temperature X of an environment. Due to measurement noise, the observed value is

$$Y = X + N,$$

where $N \sim \mathcal{N}(0, \sigma_N^2)$.

Assume that the true temperature is a random variable with

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2),$$

and that X and N are independent.

After observing $Y = y$, the receiver estimates the true temperature using the rule

$$\hat{X} = \mathbb{E}[X \mid Y = y].$$

- (a) Find the conditional distribution of $X \mid Y = y$.
- (b) Compute $\hat{X} = \mathbb{E}[X \mid Y = y]$ in terms of y , μ_X , σ_X^2 , and σ_N^2 .
- (c) Interpret the result and explain how the estimate changes as the noise variance σ_N^2 increases.

Problem 3

A communication system sends one of two possible signals, $X = 1$ or $X = -1$. Due to Gaussian noise in the channel, the received signal is

$$Y = X + N,$$

where $N \sim \mathcal{N}(0, \sigma^2/2)$.

The probabilities of transmission are

$$\Pr(X = 1) = \frac{1}{a}, \quad \Pr(X = -1) = 1 - \frac{1}{a},$$

where $a > 1$.

After observing the received value $Y = y$, the receiver decides the transmitted signal according to the following rule:

$$\hat{X} = \arg \max_{x \in \{-1, 1\}} P_{X|Y}(x | y).$$

(a) Show that the decision rule can be written as

$$\hat{X} = \arg \max_{x \in \{-1, 1\}} \Pr(X = x) f_{Y|X}(y | x).$$

(b) Show that the decision rule is equivalent to comparing

$$\Pr(X = 1)f_{Y|X}(y|1) \quad \text{and} \quad \Pr(X = -1)f_{Y|X}(y|-1).$$

(c) Show that the decision rule can be expressed using a threshold b such that

$$\hat{X} = \begin{cases} 1, & Y > b, \\ -1, & Y < b. \end{cases}$$

And, find the value of b in terms of a and σ^2 .

Problem 4

Let X and Y be jointly continuous random variables with joint PDF

$$f_{X,Y}(x, y) = \begin{cases} c(x + y), & 0 \leq x \leq 1, x^2 \leq y \leq x, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the constant c .

(b) Find the marginal PDF $f_X(x)$.

(c) Find the marginal PDF $f_Y(y)$.

(d) Find $f_{Y|X}(y|x)$, $F_{Y|X}(y|x)$, $\mathbb{E}[Y | X = x]$, $\mathbb{V}[Y | X = x]$.

Problem 5

A random variable X takes values in $\{1, 2, 3\}$. Given an observation $Y = y$, suppose the likelihoods are

$$f_{Y|X}(y|1) = e^2, \quad f_{Y|X}(y|2) = e^1, \quad f_{Y|X}(y|3) = e^{-1}.$$

Assume a uniform prior:

$$\Pr(X = 1) = \Pr(X = 2) = \Pr(X = 3) = \frac{1}{3}.$$

- (a) Use Bayes' rule to compute $\Pr(X = i | Y = y)$ for $i = 1, 2, 3$.
 (b) Show that the posterior can be written as

$$\Pr(X = i | Y = y) = \frac{e^{s_i}}{\sum_{j=1}^3 e^{s_j}},$$

for appropriate values of s_i .

- (c) Identify s_i and explain why this expression is called the softmax function.

Problem 6

The moment generating function (MGF) and the characteristic function (CF) of a random variable X are defined as

$$M_X(t) = \mathbb{E}[e^{tX}], \quad \phi_X(t) = \mathbb{E}[e^{itX}].$$

- (a) Show how the n -th moment $\mathbb{E}[X^n]$ can be obtained from the MGF $M_X(t)$.
 (b) Suppose the MGF of a random variable X is given by

$$M_X(t) = e^{3t+2t^2}.$$

Find $\mathbb{E}[X]$ and $\text{Var}(X)$.

- (c) Find the characteristic function of $X \sim \mathcal{N}(0, 1)$.
 (d) Compare the MGF and the CF in terms of their existence and their ability to uniquely determine the distribution of a random variable.

Problem 7

Let X and Y be continuous random variables with joint PDF

$$f_{X,Y}(x,y) = \begin{cases} c(x+y)e^{-x}, & 0 \leq y \leq x, 0 \leq x < \infty, \\ 0, & \text{otherwise.} \end{cases}$$

Define the event

$$A = \{X + Y \leq 2\}.$$

- (a) Find the constant c .
- (b) Find the conditional joint PDF $f_{X,Y|A}(x,y)$.
- (c) Find the conditional marginal PDF $f_{X|A}(x)$.
- (d) Find $\mathbb{E}[X | A]$, $\mathbb{V}[X | A]$.

Problem 8

Let X be a Gaussian random variable with mean $\mu = 70$ and standard deviation $\sigma = 10$, i.e.,

$$X \sim \mathcal{N}(70, 10^2).$$

- (a) Express $\Pr(X \leq 85)$ in terms of the Q -function.
- (b) Express $\Pr(60 \leq X \leq 80)$ in terms of the error function.
- (c) Express $\Pr(X > 90)$ in terms of the complementary error function.

Problem 9

A binary communication system transmits bits $B \in \{0, 1\}$ with

$$\Pr(B = 0) = \Pr(B = 1) = \frac{1}{2}.$$

Let H denote the channel power gain, and suppose that H follows an exponential distribution with parameter 1.

For a given $H = h$ and $B = b$, the bit error probability is

$$\Pr(\text{error} \mid B = b, H = h) = Q\left(\sqrt{2\rho h}\right),$$

where ρ is a constant representing the average signal-to-noise ratio.

- (a) Write the PDF $f_H(h)$ of H .
- (b) Using the law of total probability, derive a closed-form expression for $\Pr(\text{error})$.

Problem 10

Let $Z \sim \text{Uniform}(0, 1)$. Given $Z = z$, let X be a continuous random variable with conditional PDF

$$f_{X|Z}(x|z) = \begin{cases} \frac{1}{z}, & 0 \leq x \leq z, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Find $\mathbb{E}[X \mid Z = z]$.
- (b) Using the law of total expectation, compute $\mathbb{E}[X]$.
- (c) Compute $\mathbb{E}[X^2 \mid Z = z]$.
- (d) Using the law of total expectation, compute $\mathbb{E}[X^2]$.
- (e) Find $\text{Var}(X)$.

Problem 11

Let X, Y, Z be continuous random variables with joint PDF

$$f_{X,Y,Z}(x,y,z) = \begin{cases} c e^{-z}, & 0 \leq x \leq z, 0 \leq y \leq z, z \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Find the constant c .
- (b) Find the conditional joint PDF $f_{X,Y|Z}(x,y|z)$.
- (c) Determine whether X and Y are conditionally independent given Z .

Problem 12

Let Z be an exponential random variable with parameter $\lambda > 0$. Given $Z = z$, let X and Y be independent exponential random variables with parameter z , i.e.,

$$X \mid Z = z \sim \text{Exp}(z), \quad Y \mid Z = z \sim \text{Exp}(z).$$

Determine whether X and Y are independent.